

# Intro To Thermo-E26-I

 Edit

## Instructor:

- Prof. Reza Ebadi (Send emails directly. Please don't send a message via Canvas)
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## Grader:

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## Location and Time:

- The lectures will be recorded and shared via Echo360

## Office Hours:

- By appointment via Zoom. You may let me know during the lecture or via email should you need assistance with the lecture and/or assignments.

## Required Text:

- *Fundamentals of Engineering Thermodynamics*, Michale J. Moran, Howard N. Shapiro, Daisie D. Boettner, Margaret B. Bailey, Wiley (ISBN 13: 9781119721437)

## Course Objectives:

1. Apply fundamental energy concepts to thermodynamic problem setups
2. Define and distinguish thermodynamic systems, states, and processes
3. Select and interpret thermodynamic property data
4. Apply mass and energy rate balances to steady-state engineering devices
5. Explain the second law of thermodynamics and use it to identify the direction of real processes

and limitations on device performance

6. Evaluate entropy generation and apply entropy balances to assess irreversibilities in engineering systems
7. Analyze vapor-compression refrigeration and heat pump cycles to determine coefficient of performance and system capacity

### **Recommended background:**

- Basic physics (PH 1110, PH 1111)
- Elementary differential and integral calculus (MA 1021, MA 1022)

### **Grading:**

- Assignments (20%): 4-5 homework assignments (one at the end of each week).
- Quizzes (20%): 4 quizzes
- Exams (60%): 2 exams

### **AI Use Policy:**

**I cannot monitor how you use AI outside of assessments (I don't even want to), But I do expect you to understand the material.**

- **Allowed:**

- Conceptual understanding: definitions, comparisons, intuition building
- Setting up problem frameworks and identifying governing equations
- Checking your reasoning after you've attempted a problem

- **Not Allowed:**

- AI use during quizzes or exams (unless explicitly allowed)
- Submitting AI-generated solutions as your own work
- Citing AI as a source

- **Use with Caution (AI is unreliable for):**

- Thermodynamic property data (steam tables, refrigerant tables, air tables): it will hallucinate values
- Data interpolation: AI gets this wrong consistently
- Numerical cycle analysis: errors in state points cascade through the entire solution

- **Thermodynamics is foundational** for what comes next: heat transfer, fluid mechanics, heat

transfer, and many MQPs.

- **Your future employers expect you to verify AI**, not trust it blindly. Engineers are responsible for their work. "The AI told me" is not a professional defense.
- **AI makes mistakes**. It will confidently give you wrong answers. If you don't understand the underlying concepts, you won't catch them.
- **How to use AI without sabotaging yourself:**
  - Attempt problems first. Struggle is where learning happens.
  - If stuck, ask AI to explain concepts - not solve the problem for you.
  - If you use AI to check your work, make sure you can explain every step.
  - If you can't explain it, you didn't learn it.
- **You can outsource your thinking to AI, you cannot outsource your understanding.**

- **Final Grade Letter:**

- A:  $\geq 90$
- B:  $\geq 80$
- C:  $\geq 70$
- NR:  $<70$

## Course Policies and Notes:

- Homework assignments, quizzes, and exams must be scanned, saved as PDF, and submitted to Canvas (no hard copies). If you have a smart phone you may use free applications such as **Genius Scan** or **MS OfficeLens**, among others to do so.
- Late homework and labs will not be accepted for any reason unless you have made prior arrangements with the instructor.
- I tend to publish the solution shortly after the deadline. Therefore, I cannot accept late submission once the solution is published.
- Discussing the problems with a classmate is allowed and encouraged, but the homework assignments, quizzes and exams must be done individually.

## **Academic Honesty:**

- Ethical behavior is essential for the engineering professionals due to the nature of the work and is expected during your academic training as well. You are required to comply with all **University policies** (<https://www.wpi.edu/sites/default/files/docs/Offices/Marketing-Communications/AIStudentguide2019-2020-pages.pdf>) regarding Academic Honesty and to familiarize yourself with those policies.
- Collaborating on assignments is acceptable in this course to support peer-to-peer learning. HOWEVER, you should only discuss the problems conceptually and refrain from exchanging digital copies of the assignments. Let me know if you have any questions about what constitutes academic dishonesty in this course.

## **Accessibility Services for Students:**

- WPI is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must contact the Office of **Accessibility Services** (<https://www.wpi.edu/offices/office-accessibility-service>) and get them send me proper accommodation letters.

## **Emotional or Mental Health Distress:**

Your academic success in this course is very important to me. If, during the term, you find emotional or mental health issues are affecting that success, please contact WPI's **Student Development & Counseling Center** (<https://www.wpi.edu/offices/student-development-counseling-center>), which provides counseling appointments and other mental health services.

## **Netiquette Guidelines:**

- Netiquette is the socially and professionally acceptable way to communicate on the Internet. Please abide by these guidelines of "netiquette" when using online communication tools with your classmates and instructor.
- In discussion boards and email messages:
  - Identify yourself. Begin messages with a greeting and close with your name.
  - Avoid sarcasm. It can be misinterpreted and cause hurt feelings.
  - Keep the dialogue collegial and professional. Some discussion topics may be controversial.
  - Do not flame -These are outbursts of extreme emotion or opinion. Think twice before you submit a response. You cannot edit or delete your posts once they have been submitted.

- Do not use offensive language or profanity.
- Use clear subject lines for your posts.
- Do not use all caps. It is the online equivalent of YELLING!
- Be forgiving. Anyone can make a mistake.
- In Zoom or video conferences:
  - Turn on your camera when your network bandwidth and learning space allows. Facial expressions and body language are an important part of communicating.
  - Connect a few minutes early.
  - Remove clutter or personal items around you.
  - Avoid background noise.
  - Consider using a headset to reduce distractions.
  - Keep your device (phone, computer, etc.) on mute unless you are speaking.
  - Speak clearly, but not too loudly.
  - Don't abuse the chat box, keep the conversation respectful and on topic.
  - Remember that a video conference has the same degree of respect as a live class, consider your appearance behavior on camera the same you would in the classroom.

## **Tentative Course Outline:**

Here is a tentative class schedule. **Please note the dates/topics are subject to change.**

### Week 0:

- Course introductions

### Week 1:

- Introduction to Thermodynamics
- Energy - Work
- Heat - Cycles
- Evaluating Properties

### Week 2:

- Specific Heat - Ideal Gas
- Control Volume

- Common Engineering Devices
- System Integration

#### Week 3:

- Second law of Thermodynamics
- Maximum Performance - Carnot Cycle
- Entropy Introduction
- Entropy Balance
- Midterm Review

#### Week 4:

- **Midterm Exam**
- T-dS Equations
- Isentropic Processes
- Vapor Power Systems - Rankine Cycle

#### Week 5:

- Rankine Cycle Modifications
- Vapor Refrigeration Systems
- Final Review
- **Final Exam**